2 Installation

2.1 Overview

Introduction

Installation of the UDC3200 consists of mounting and wiring the controller according to the instructions given in this section. Read the pre-installation information, check the model number interpretation (Subsection 2.3), and become familiar with your model selections, then proceed with installation.

What's in this section?

The following topics are covered in this section.

Pre-installation Information

If the controller has not been removed from its shipping carton, inspect the carton for damage then remove the controller.

- Inspect the unit for any obvious shipping damage and report any damage due to transit to the carrier.
- Make sure a bag containing mounting hardware is included in the carton with the controller.
- Check that the model number shown on the inside of the case agrees with what you have ordered.

2.2 Condensed Specifications

Honeywell recommends that you review and adhere to the operating limits listed in Table 2-1 when you install your controller.

Table 2-1 Condensed Specifications

Specifications	
Analog Inputs	*Accuracy:* ± 0.20% of full scale typical (± 1 digit for display) Can be field calibrated to ± 0.05% of full scale typical 16-bit resolution typical *Sampling* Rate: Both inputs are sampled six times per second *Temperature Stability:* ± 0.01% of Full Scale span / ˚C change—typical *Input Impedance:* 4-20 Milliampere Input: 250 ohms 0-10 Volt Input: 200K ohms All Others: 10 megohms *Maximum Lead Wire Resistance:* Thermocouples: 50 ohms/leg 100 ohm, 200 ohm and 500 ohm RTD: 100 ohms/leg 100 ohm Low RTD: 10 ohms/leg *Slidewire Inputs for Position Proportional Control:* 100 ohm minimum to 1000 ohms maximum
Analog Input Signal Failure Operation	*Burnout Selections:* Upscale, Downscale, Failsafe or None *Thermocouple Health:* Good, Failing, Failure Imminent or Failed *Failsafe Output Level:* Configurable 0-100% of Output range
Stray Rejection	**Common Mode** *AC (50 or 60 Hz):* 120 dB (with maximum source impedance of 100 ohms) or ± 1 LSB (least significant bit) whichever is greater with line voltage applied. *DC:* 120 dB (with maximum source impedance of 100 ohms) or a ±1 LSB whichever is greater with 120 Vdc applied. *DC (to 1 KHz):* 80 dB (with maximum source of impedance of 100 ohms) or ±1 LSB whichever is greater with 50 Vac applied. **Normal Mode** *AC (50 or 60 Hz):* 60 dB (with 100 % span peak-to-peak maximum)
Digital Inputs (Two) (Optional)	+30 Vdc source for external dry contacts or isolated solid state contacts. Digital Inputs are isolated from line power, earth ground, analog inputs and all outputs except for the Second Current Output. The second Digital Input is mutually exclusive with the Second Current Output.

Specifications	
Controller Output Types	**Electromechanical Relays (One or Two)** SPDT contacts. Both Normally Open and Normally Closed contacts are brought out to the rear terminals. Internally socketed. *Resistive Load:* 5 amps @ 120 Vac or 240 Vac or 30 Vdc *Inductive Load (cosφ = 0.4):* 3 amps @ 130 Vac or 250 Vac *Inductive Load (L/R = 7 msec):* 3.5 amps @ 30 Vdc *Motor:* 1/6 H.P. **Dual Electromechanical Relays** Two SPST relays. One Normally Closed contact for each relay is brought out to the rear terminals. This option takes the place of one of the above electromechanical relays, and is especially useful for Time Duplex or Three Position Step Control or Position Proportional Control applications. Instruments with this option can have a total of 4 relays plus one current output. Internally socketed *Resistive Load:* 2 amps @ 120 Vac, 240 Vac or 30 Vdc *Inductive Load (cosφ = 0.4):* 1 amp @ 130 Vac or 250 Vac *Inductive Load (L/R = 7 msec):* 1 amp @ 30 Vdc **Solid State Relays (One or Two)** Zero-crossing type SPST solid state contacts consisting of a triac N.O. output. Internally socketed. *Resistive Load:* 1.0 amp @ 25°C and 120 or 240 Vac, 0.5 amp @ 55°C and 120 or 240 Vac *Inductive Load:* 50 VA @ 120 Vac or 240 Vac *Minimum Load:* 20 milliamps **Open Collector Outputs (One or Two)** Socketed assembly replacing a relay. Opto-isolated from all other circuits except current output and not from each other. Internally powered @ 30 Vdc. Note: Applying an external power supply to this output will damage the instrument. *Maximum Sink Current:* 20 mA *Short-circuit current limit:* 100 mA **Current Outputs (One or Two)** These outputs provide a 21 mA dc maximum into a negative or positive grounded load or into a non-grounded load. Current outputs are isolated from each other, line power, earth ground and all inputs. Outputs can be easily configured via the keyboard for either direct or reverse action and for either 0 to 20 mA or 4 to 20 mA without field calibration. Both current outputs can be used in an Auxiliary Output mode. This Auxiliary Output can be configured to represent Input, PV, Setpoint, Deviation, or Control output. The range of an Auxiliary Output can be scaled per the range of the selected variable and can be set anywhere between 0 to 21 mA. The Second Current Output is mutually exclusive with the second Digital Input. *Resolution:* 14 bits over 0 to 21 mA *Accuracy:* 0.05% of full scale *Temperature Stability:* 0.01% F.S./°C *Load Resistance:* 0 to 1000 ohms
Alarm Outputs (Optional)	One SPDT Electromechanical relay. A second alarm is available if the second control relay is not used for control purposes or when the Dual Relay Option is used. Up to four setpoints are independently set as high or low alarm, two for each relay. Setpoint can be on any Input, Process Variable, Deviation, Manual Mode, Failsafe, PV Rate, RSP Mode, Communication Shed, or Output. A single adjustable hysteresis of 0.0 to 100.0% is provided. The alarm can also be set as an ON or OFF event at the beginning of a Setpoint ramp/soak segment. *Alarm Relay Contacts Rating:* Resistive Load: 5 amps at 120 Vac or 240 Vac or 30 Vdc
Isolation (Functional)	*AC Power:* Electrically isolated from all other inputs and outputs and earth ground to withstand a HIPOT potential of 1900 Vdc for 2 seconds per Annex K of EN61010-1. *Analog Inputs and Outputs:* Are isolated from each other and all other circuits at 850 Vdc for 2 seconds. *Digital Inputs and Digital Outputs:* Electrically isolated from all other circuits to withstand a HIPOT potential of 850 Vdc for 2 seconds per Annex K of EN61010-1.

Specifications	
	Relay Contacts: With a working voltage of 115/230 Vac, these are electrically isolated from all other circuits to withstand a HIPOT potential of 345 Vdc for 2 seconds per Annex K of EN61010-1
RS422/485 Modbus RTU Communications Interface (Optional)	*Baud Rate:* 4800, 9600,19,200 or 38,400 baud selectable *Data Format:* Floating point or integer *Length of Link:* 2000 ft (600 m) max. with Belden 9271 Twinax Cable and 120 ohm termination resistors 4000 ft. (1200 m) max. with Belden 8227 Twinax Cable and 100 ohm termination resistors *Link Characteristics:* Two-wire, multi-drop Modbus RTU protocol, 15 drops maximum or up to 31 drops for shorter link length.
Ethernet TCP/IP Communications Interface (Optional)	*Type:* 10Base-T *Length of Link:* 330 ft. (100 m) maximum *Link Characteristics:* Four-wire, single drop, five hops maximum *IP Address:* IP Address is 10.0.0.2 as shipped from the Factory *Recommended network configuration:* Use Switch rather than Hub in order to maximize UDC Ethernet performance
Infrared Communications (Standard)	*Type:* Serial Infrared (SIR) *Length of Link:* 3 ft. (1 m) maximum for IrDA 1.0 compliant devices *Baud Rate:* 19,200 or 38,400 baud selectable
Power Consumption	20 VA maximum (90 to 264 Vac) 15 VA maximum (24 Vac/dc)
Power Inrush Current	10A maximum for 4 ms (under operating conditions), reducing to a maximum of 225 mA (90 to 264 Vac operation) or 750 mA (24 Vac/dc operation) after one second. CAUTION When applying power to more than one instrument, make sure that sufficient power is supplied. Otherwise, the instruments may not start up normally due to voltage drop from the inrush current.
Weight	3 lbs. (1.3 kg)

Environmental and Operating Conditions

Parameter	Reference	Rated	Operative Limits	Transportation and Storage
Ambient Temperature	25 ± 3 °C 77 ± 5 °F	15 to 55 °C 58 to 131 °F	0 to 55 °C 32 to 131 °F	–40 to 66 °C –40 to 151 °F
Relative Humidity	10 to 55*	10 to 90*	5 to 90*	5 to 95*
Vibration Frequency (Hz) Acceleration (g)	 0 0	 0 to 70 0.4	 0 to 200 0.6	 0 to 200 0.5
Mechanical Shock Acceleration (g) Duration (ms))	 0 0	 1 30	 5 30	 20 30
Line Voltage (Vdc)	+24 ± 1	22 to 27	20 to 30	- -
Line Voltage (Vac) **90 to 240 Vac** **24 Vac**	 120 ± 1 240 ± 2 24 ± 1	 90 to 240 20 to 27	 90 to 264 20 to 27	 - - - - - -
Frequency (Hz) (For Vac)	50 ± 0.2 60 ± 0.2	49 to 51 59 to 61	48 to 52 58 to 62	- - - -

* The maximum moisture rating only applies up to 40 °C (104 °F). For higher temperatures, the RH specification is derated to maintain constant moisture content.

2.3 Model Number Interpretation

Introduction

Write your controller's model number in the spaces provided below and circle the corresponding items in each table. This information will also be useful when you wire your controller.

Instructions

- Select the desired key number. The arrow to the right marks the selection available.
- Make the desired selections from Tables I through VI using the column below the proper arrow. A dot (•) denotes availability.

Key Number		I		II		III		IV		V		VI
_ _ _ _ _ _ _	-	_ _	-	_ _ _ _	-	_ _ _	-	_ _ _ _ _ _	-	_ _	-	_

KEY NUMBER - UDC3200 Single Loop Controller

Description	Selection	Availability	
Digital Controller for use with 90 to 264Vac Power	DC3200	↓	
Digital Controller for use with 24Vac/dc Power	DC3201		↓

TABLE I - Specify Control Output and/or Alarms

Output #1	Current Output (4 to 20ma, 0 to 20 ma)	C _	•	•
	Electro Mechanical Relay (5 Amp Form C)	E _	•	•
	Solid State Relay (1 Amp)	A _	•	•
	Open Collector transistor output	T _	•	•
	Dual 2 Amp Relays (Both are Form A) (Heat/Cool Applications)	R _	•	•
Output #2 and Alarm #1 or Alarms 1 and 2	No Additional Outputs or Alarms	_ 0	•	•
	One Alarm Relay Only	_ B	•	•
	E-M Relay (5 Amp Form C) Plus Alarm 1 (5 Amp Form C Relay)	_ E	•	•
	Solid State Relay (1 Amp) Plus Alarm 1 (5 Amp Form C Relay)	_ A	•	•
	Open Collector Plus Alarm 1 (5 Amp Form C Relay)	_ T	•	•

TABLE II - Communications and Software Selections

Communications	None	0 _ _ _	•	•
	Auxiliary Output/Digital Inputs (1 Aux and 1 DI or 2 DI)	1 _ _ _	•	•
	RS-485 Modbus Plus Auxiliary Output/Digital Inputs	2 _ _ _	•	•
	10 Base-T Ethernet (Modbus RTU) Plus Auxiliary Output/Digital Inputs	3 _ _ _	•	•
Software Selections	Standard Functions, Includes Accutune	_ 0 _ _	•	•
	Math Option	_ A _ _	•	•
	Set Point Programming (1 Program, 12 Segments)	_ B _ _	•	•
	Set Point Programming Plus Math	_ C _ _	•	•
Reserved	No Selection	_ _ 0 _	•	•
Infrared interface	Infrared Interface Included (Can be used with a Pocket PC)	_ _ _ R	•	•

TABLE III - Input 1 can be changed in the field using external resistors

Input 1	TC, RTD, mV, 0-5V, 1-5V	1 _ _	•	•
	TC, RTD, mV, 0-5V, 1-5V, 0-20mA, 4-20mA	2 _ _	•	•
	TC, RTD, mV, 0-5V, 1-5V, 0-20mA, 4-20mA, 0-10V	3 _ _	•	•
	Carbon, Oxygen or Dewpoint (Requires Input 2)	1 6 0	•	•
Input 2	None	_ 00	•	•
	TC, RTD, mV, 0-5V, 1-5V, 0-20mA, 4-20mA	_ 10	•	•
	TC, RTD, mV, 0-5V, 1-5V, 0-20mA, 4-20mA, 0-10V	_ 20	•	•
	Slidewire Input (Requires two Relay Outputs)	_ 40	•	•

TABLE IV - Options

Approvals	CE (Standard)	0 _ _ _ _	•	•
	CE, UL and CSA	1 _ _ _ _	•	•
Tags	None	_ 0 _ _ _	•	•
	Linen Customer ID Tag - 3 lines w/22 characters/line	_ T _ _ _	•	•
	Stainless Steel Customer ID Tag - 3 lines w/22 characters/line	_ S _ _ _	•	•
Future Options	None	_ _ 0 _ _	•	•
	None	_ _ _ 0 _	•	•
	None	_ _ _ _ 0	•	•

TABLE V - Product Manuals

Manuals	Product Information on CD - All Languages	0 _	•	•
	English Manual	E _	•	•
	French Manual	F _	•	•
	German Manual	G _	•	•
	Italian Manual	I _	•	•
	Spanish Manual	S _	•	•
Certificate	None	_ 0	•	•
	Certificate of Conformance (F3391)	_ C	•	•

TABLE VI

No Selection	None	0 _	•	•

Figure 2-1 Model Number Interpretation

2.4 Control and Alarm Relay Contact Information

Control Relays

ATTENTION

Control relays operate in the standard control mode (that is, energized when output state is on).

Table 2-2 Control Relay Contact Information

Unit Power	Control Relay Wiring	Control Relay Contact	Output #1 or #2 Indicator Status
Off	N.O.	Open	**Off**
	N.C.	Closed	
On	N.O.	Open	**Off**
		Closed	**On**
	N.C.	Closed	**Off**
		Open	**On**

Alarm Relays

ATTENTION

Alarm relays are designed to operate in a failsafe mode (that is, de-energized during alarm sate). This results in alarm actuation when power is OFF or when initially applied, until the unit completes self-diagnostics. If power is lost to the unit, the alarms will de-energize and thus the alarm contacts will close.

Table 2-3 Alarm Relay Contact Information

Unit Power	Alarm Relay Wiring	Variable NOT in Alarm State		Variable in Alarm State	
		Relay Contact	Indicators	Relay Contact	Indicators
Off	N.O.	Open	**Off**	Open	**Off**
	N.C.	Closed		Closed	
On	N.O.	Closed	**Off**	Open	**On**
	N.C.	Open		Closed	

2.5 Mounting

Physical Considerations

The controller can be mounted on either a vertical or tilted panel using the mounting kit supplied. Adequate access space must be available at the back of the panel for installation and servicing activities.

- Overall dimensions and panel cutout requirements for mounting the controller are shown in Figure 2-2.
- The controller's mounting enclosure must be grounded according to CSA standard C22.2 No. 0.4 or Factory Mutual Class No. 3820 paragraph 6.1.5.
- The front panel is moisture rated NEMA3 and IP55 rated and can be easily upgraded to NEMA4X and IP66. (See Mounting Method, page 19)

Overall Dimensions

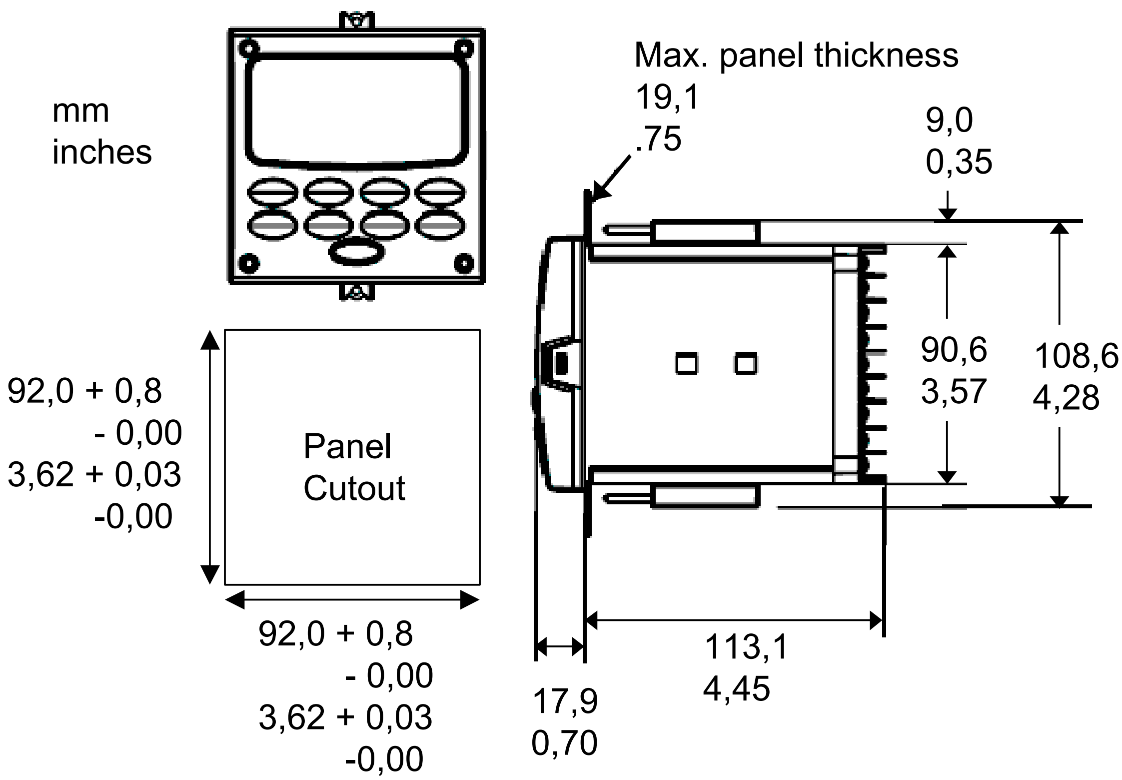


Figure 2-2 Mounting Dimensions (not to scale)

Mounting Notes

Before mounting the controller, refer to the nameplate on the outside of the case and make a note of the model number. It will help later when selecting the proper wiring configuration.

Mounting Method

Before mounting the controller, refer to the nameplate on the outside of the case and make a note of the model number. It will help later when selecting the proper wiring configuration.

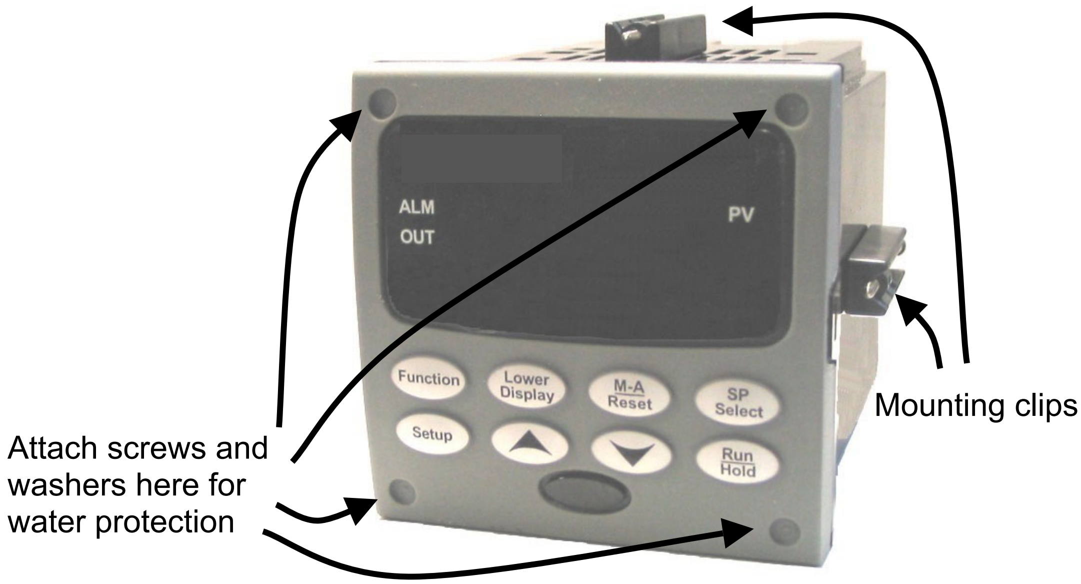


Figure 2-3 Mounting Methods

Mounting Procedure

Table 2-4 Mounting Procedure

Step	Action
1	Mark and cut out the controller hole in the panel according to the dimension information in Figure 2-2.
2	Orient the case properly and slide it through the panel hole from the front.
3	Remove the mounting kit from the shipping container and install the kit as follows: • For **normal (NEMA 3/IP55) installation two mounting clips** are required. Insert the prongs of the clips into the two holes in the top and bottom center of the case • For **water-protected (NEMA 4X/IP66) installation four mounting clips** are required. There are two options of where to install the mounting clips: 1) Insert the prongs of the clips into the two holes on the left and right side of the top and bottom of the case or 2) on the center on each of the four sides. • Tighten screws to 2 lb-inch (22 N•cm) to secure the case against the panel. CAUTION: Over tightening will cause distortion and the unit may not seal properly.
4	For water-protected installation, install four screws with washers into the four recessed areas in the corners of the front bezel (Figure 2-3). Push the point of the screw through the center piercing the elastomeric material and then tighten screws to 5 lb-in (56 N•cm).

2.6 Wiring

2.6.1 Electrical Considerations

Line voltage wiring

This controller is considered “rack and panel mounted equipment” per EN61010-1, Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use, Part 1: General Requirements. Conformity with 72/23/EEC, the Low Voltage Directive requires the user to provide adequate protection against a shock hazard. The user shall install this controller in an enclosure that limits OPERATOR access to the rear terminals.

Mains Power Supply

This equipment is suitable for connection to 90 to 264 Vac or to 24 Vac/dc 50/60 Hz, power supply mains. It is the user’s responsibility to provide a switch and non-time delay (North America), quick-acting, high breaking capacity, Type F (Europe), 1/2A, 250V fuse(s), or circuit-breaker for 90-264 Vac applications; or 1 A, 125 V fuse or circuit breaker for 24 Vac/dc applications, as part of the installation. The switch or circuit-breaker shall be located in close proximity to the controller, *within easy reach of the OPERATOR*. The switch or circuit-breaker shall be marked as the disconnecting device for the controller.

CAUTION

Applying 90-264 Vac to an instrument rated for 24 Vac/dc will severely damage the instrument and is a fire and smoke hazard.

When applying power to multiple instruments, make certain that sufficient current is supplied. Otherwise, the instruments may not start up normally due to the voltage drop caused by the in-rush current.

Controller Grounding

PROTECTIVE BONDING (grounding) of this controller and the enclosure in which it is installed shall be in accordance with National and Local electrical codes. To minimize electrical noise and transients that may adversely affect the system, supplementary bonding of the controller enclosure to a local ground, using a No. 12 (4 mm^2) copper conductor, is recommended.

Control/Alarm Circuit Wiring

The insulation of wires connected to the Control/Alarm terminals shall be rated for the highest voltage involved. Extra Low Voltage (ELV) wiring (input, current output, and low voltage Control/Alarm circuits) shall be separated from HAZARDOUS LIVE (>30 Vac, 42.4 Vpeak, or 60 Vdc) wiring per Permissible Wiring Bundling, Table 2-5.

Electrical Noise Precautions

Electrical noise is composed of unabated electrical signals which produce undesirable effects in measurements and control circuits.

Digital equipment is especially sensitive to the effects of electrical noise. Your controller has built-in circuits to reduce the effect of electrical noise from various sources. If there is a need to further reduce these effects:

- *Separate External Wiring*—Separate connecting wires into bundles (See Permissible Wiring Bundling - Table 2-5) and route the individual bundles through separate conduit metal trays.
  Use Suppression Devices—For additional noise protection, you may want to add suppression devices at the external source. Appropriate suppression devices are commercially available.

ATTENTION

For additional noise information, refer to document number *51-52-05-01, How to Apply Digital Instrumentation in Severe Electrical Noise Environments.*

Permissible Wiring Bundling

Table 2-5 Permissible Wiring Bundling

Bundle No.	Wire Functions
1	• Line power wiring • Earth ground wiring • Line voltage control relay output wiring • Line voltage alarm wiring
2	**Analog** signal wire, such as: • Input signal wire (thermocouple, 4 to 20 mA, etc.) • 4-20 mA output signal wiring **Digital** input signals
3	• Low voltage alarm relay output wiring • Low voltage wiring to solid state type control circuits • Low voltage wiring to open collector type control circuits

2.7 Wiring Diagrams

Identify Your Wiring Requirements

To determine the appropriate diagrams for wiring your controller, refer to the model number interpretation in this section. The model number of the controller is on the outside of the case.

Universal Output Functionality and Restrictions

Instruments with multiple outputs can be configured to perform a variety of output types and alarms. For example, an instrument with a current output and two relays can be configured to perform any of the following:
1) Current Simplex with two alarm relays;
2) Current Duplex 100% with two alarm relays;
3) Time Simplex with one alarm relay;
4) Time Duplex with no alarm relays; or
5) Three Position Step Control with no alarm relays.
These selections may all be made via the keyboard and by wiring to the appropriate output terminals; there are no internal jumpers or switches to change. This flexibility allows a customer to stock a single instrument which is able to handle a variety of applications.

Table 2-6 shows what control types and alarms are available based upon the installed outputs. In this table, when Duplex Control and Reverse Action are configured, “Output 1” is HEAT while “Output 2” is COOL. When Three Position Step Control is configured, “Output 1” is OPEN while “Output 2” is CLOSE. The Output 1/2 option “Single Relay” can be any of the following selections: Electro-Mechanical Relay, Solid-State Relay or Open Collector Output.

Table 2-6 Universal Output Functionality and Restrictions

Output Algorithm Type	Output 1/2 Option	Function of Output 1/2	Function of Other Outputs		
			Output #3	Output #4	Auxiliary Output
Time Simplex	Single Relay	Output 1	Alarm 2	Alarm 1	Not Needed
	Current Output	INU	Output 1	Alarm 1	Not Needed
	Dual Relay	Output 1	Alarm 2	Alarm 1	Not Needed
Time Duplex or TPSC or Position Proportional	Single Relay	Output 1	Output 2	Alarm 1	Not Needed
	Current Output	INU	Output 2	Output 1	Not Needed
	Dual Relay	Outputs 1 and 2	Alarm 2	Alarm 1	Not Needed
Current Simplex	Single Relay	INU	Alarm 2	Alarm 1	Output 1
	Current Output	Output 1	Alarm 2	Alarm 1	Not Needed
	Dual Relay	INU	Alarm 2	Alarm 1	Output 1
Current Dup. 100% Current = COOL and HEAT	Single Relay	INU	Alarm 2	Alarm 1	Outputs 1 and 2
	Current Output	Outputs 1 and 2	Alarm 2	Alarm 1	Not Needed
	Dual Relay	INU	Alarm 2	Alarm 1	Outputs 1 and 2
Current Duplex 50% Current = HEAT Aux Out = COOL	Single Relay	N/A	N/A	N/A	N/A
	Current Output	Output 1	Alarm 2	Alarm 1	Output 2
	Dual Relay	N/A	N/A	N/A	N/A
Current/Time Current = COOL Time = HEAT	Single Relay *	Output 1	Output 2	Alarm 1	Output 2
	Current Output	Output 2	Output 2	Alarm 1	Not Needed
	Dual Relay *	Outputs 1 & 2	Alarm 2	Alarm 1	Output 2
Time/Current Time = COOL Current = HEAT	Single Relay *	Output 1	Output 2	Alarm 1	Output 1
	Current Output	Output 1	Output 2	Alarm 1	Not Needed
	Dual Relay *	Outputs 1 & 2	Alarm 2	Alarm 1	Output 1

TPSC = Three Position Step Control

N/A = Not Available – This output algorithm type cannot be performed with this Output 1/2 option.

INU = Installed, Not Used – The installed Output 1/2 option is not used for the configured output algorithm type.

Not Needed = Auxiliary Output is Not Needed to provide the desired output algorithm and can be used for another purpose. With the proper configuration, Auxiliary Output could also be used as a substitute for the Current Output.

* To obtain this output algorithm type with these Output 1/2 Options: 1) Configure the OUTALG selection as “TIME D”; 2) Configure Auxiliary Output for “OUTPUT” and; 3) Scale the Auxiliary Output as necessary for the desired output algorithm type. For these

selections, the Output 1 (HEAT) and Output 2 (COOL) signals will be present both on the Auxiliary Output and on the two relays normally used for Time Duplex.

Wiring the Controller

Using the information contained in the model number, select the appropriate wiring diagrams from the composite wiring diagram below. Refer to the individual diagrams listed to wire the controller according to your requirements.

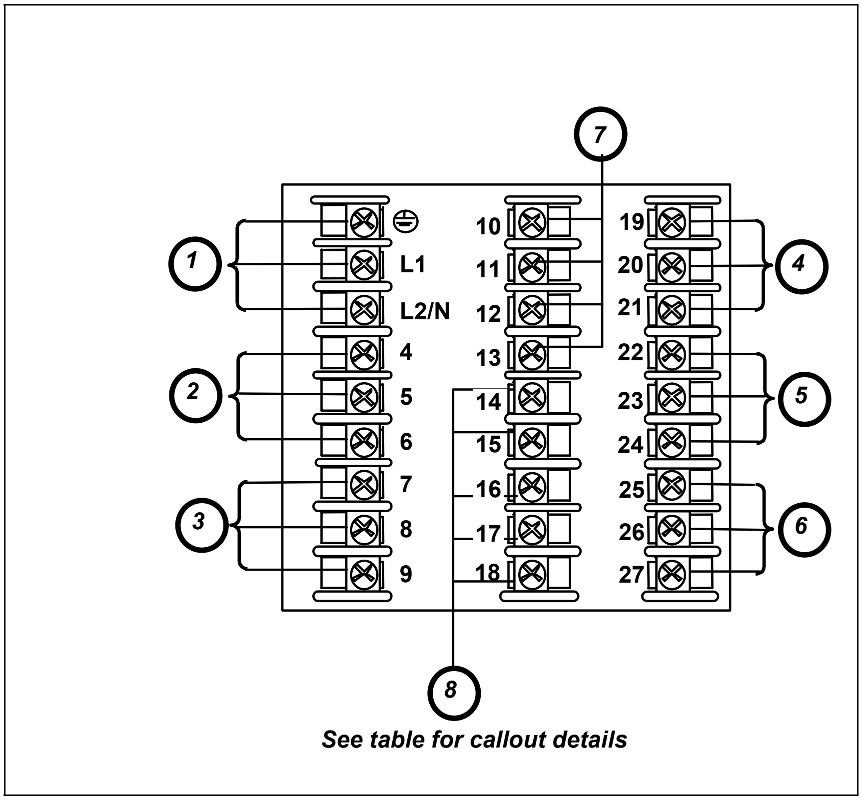
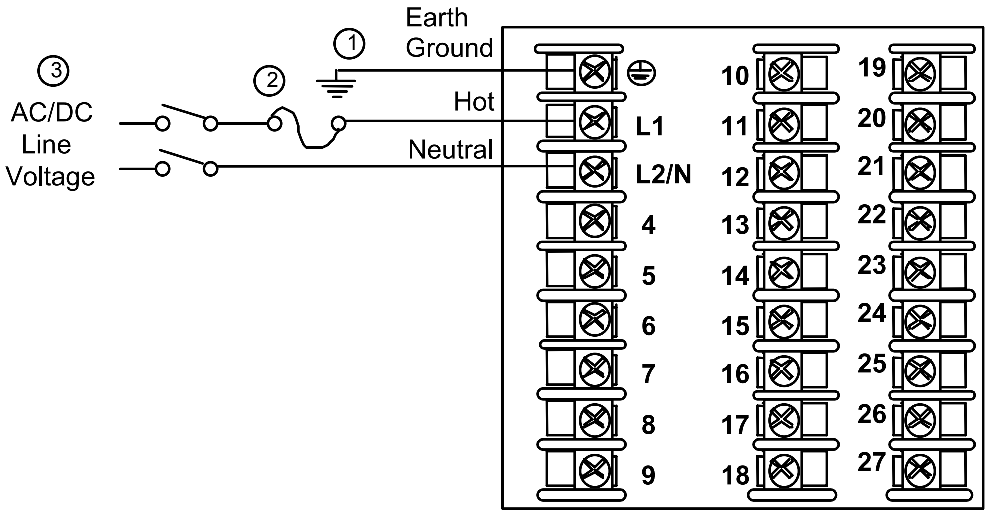


Figure 2-4 Composite Wiring Diagram

Callout	Details
1	AC/DC Line Voltage Terminals. See Figure 2-5.
2	Output 3 Terminals. See Figure 2-8 through Figure 2-14.
3	Output 4 Terminals. See Figure 2-8 through Figure 2-14.
4	Outputs 1 and 2 Terminals. See Figure 2-8 through Figure 2-14.
5	Input #2 Terminals. See Figure 2-7.
6	Input #1 Terminals. See Figure 2-6.
7	Aux. Output and Digital Inputs Terminals. See Figure 2-17.
8	Communications Terminals. See Figure 2-15 and Figure 2-16.



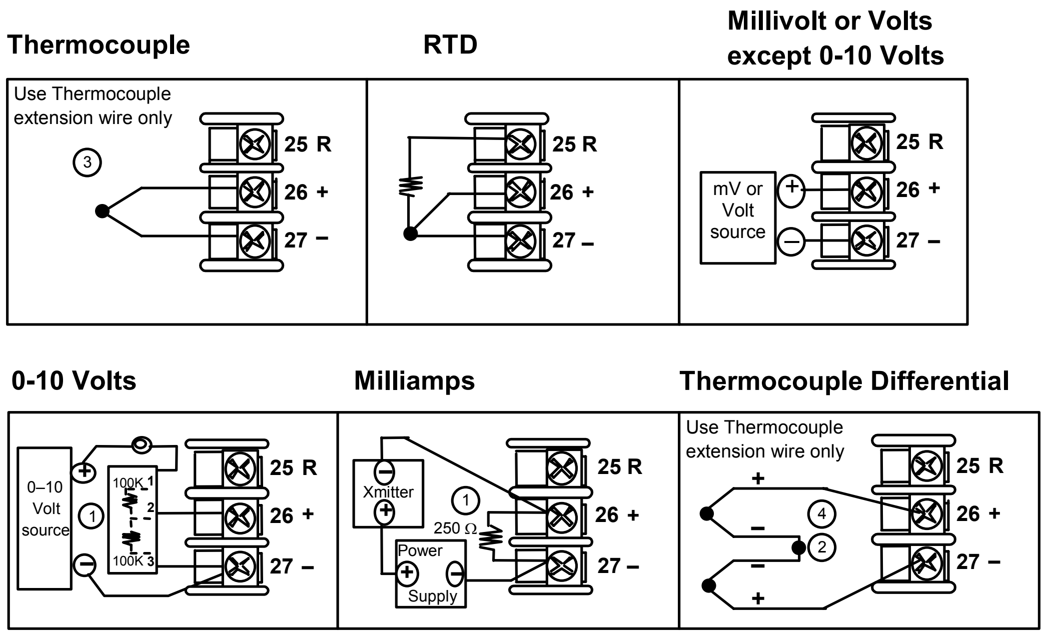
① PROTECTIVE BONDING (grounding) of this controller and the enclosure in which it is installed, shall be in accordance with National and local electrical codes. To minimize electrical noise and transients that may adversely affect the system, supplementary bonding of the controller enclosure to local ground using a No. 12 (4 mm^2) copper conductor is recommended. Before powering the controller, see "Prelimnary Checks" in this section of the Product Manual.

② It is the user's responsibility to provide a switch and non-time delay (North America), quick-acting, high breaking capacity, Type F (Europe), 1/2A, 250V fuse(s), or circuit-breaker for 90-264 Vac applications; or 1 A, 125 V fuse or circuit breaker for 24 Vac/dc applications, as part of the installation.

③ **CAUTION** Applying 90-264 Vac to an instrument rated for 24 Vac/dc will severely damage the instrument and is a fire and smoke hazard.

Figure 2-5 Mains Power Supply

Input #1



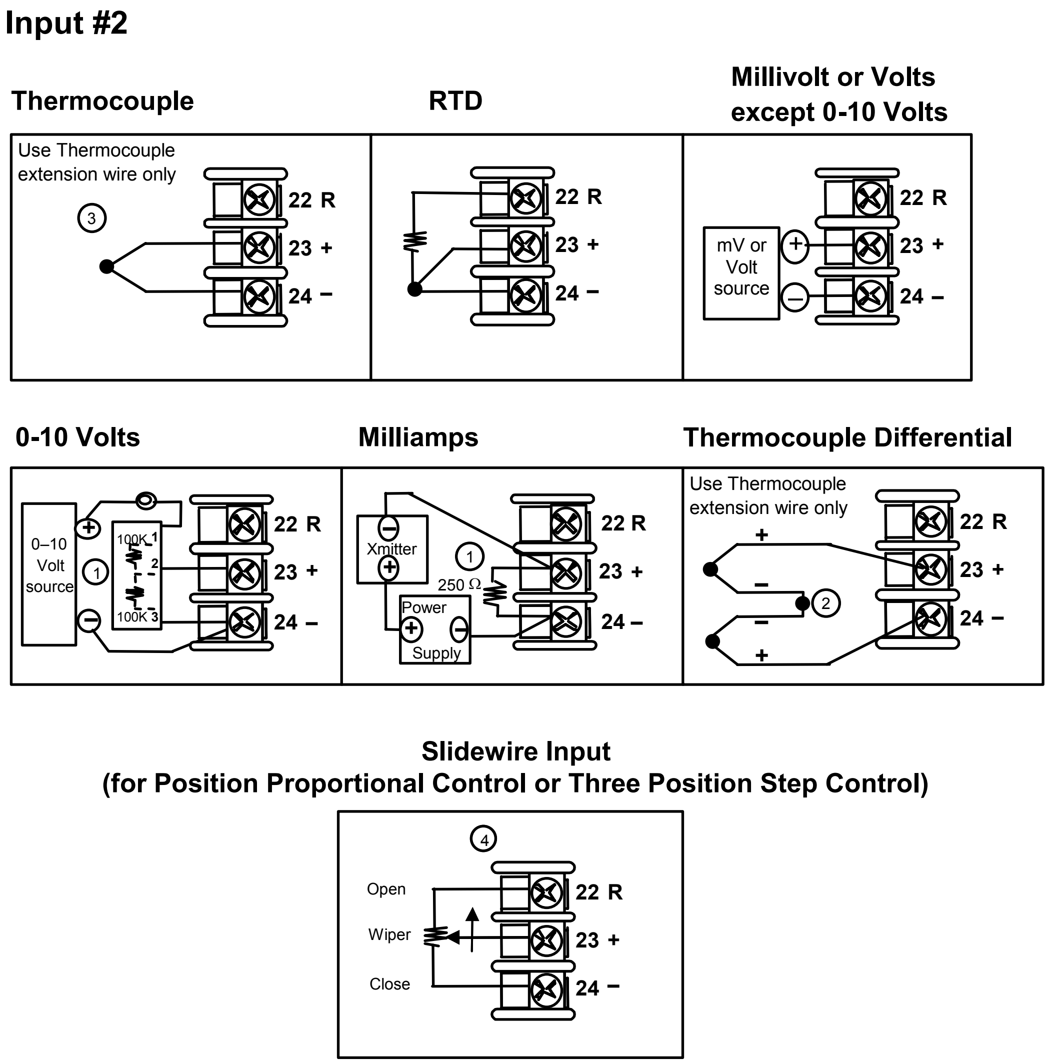
① The 250 ohm resistor for milliamp inputs or the voltage divider for 0-10 Volt inputs are supplied with the controller when those inputs are specified. These items must be installed prior to start up when the controller is wired. For 0-20 mA applications, the resistor should be located at the transmitter terminals if Burnout detection is desired.

② Splice and tape this junction between the two thermocouples. This junction may be located anywhere between the thermocouples and the instrument terminals, it does not need to be close to the other thermocouple junctions. Both thermocouples must be of the same type. For best accuracy, the two thermocouples should be matched or, preferably, made from the same batch of wire.

③ This controller does not produce a steady current for burnout detection. For that reason, when a thermocouple is used in parallel with another instrument, it may be desirable to configure the burnout selection for this controller to "NOFS" and use the burnout current from the other instrument to also drive this controller.

④ The millivolt values for the Thermocouple Differential Input are for a pair of J thermocouples at an ambient temperature mean of 450°F / 232°C.

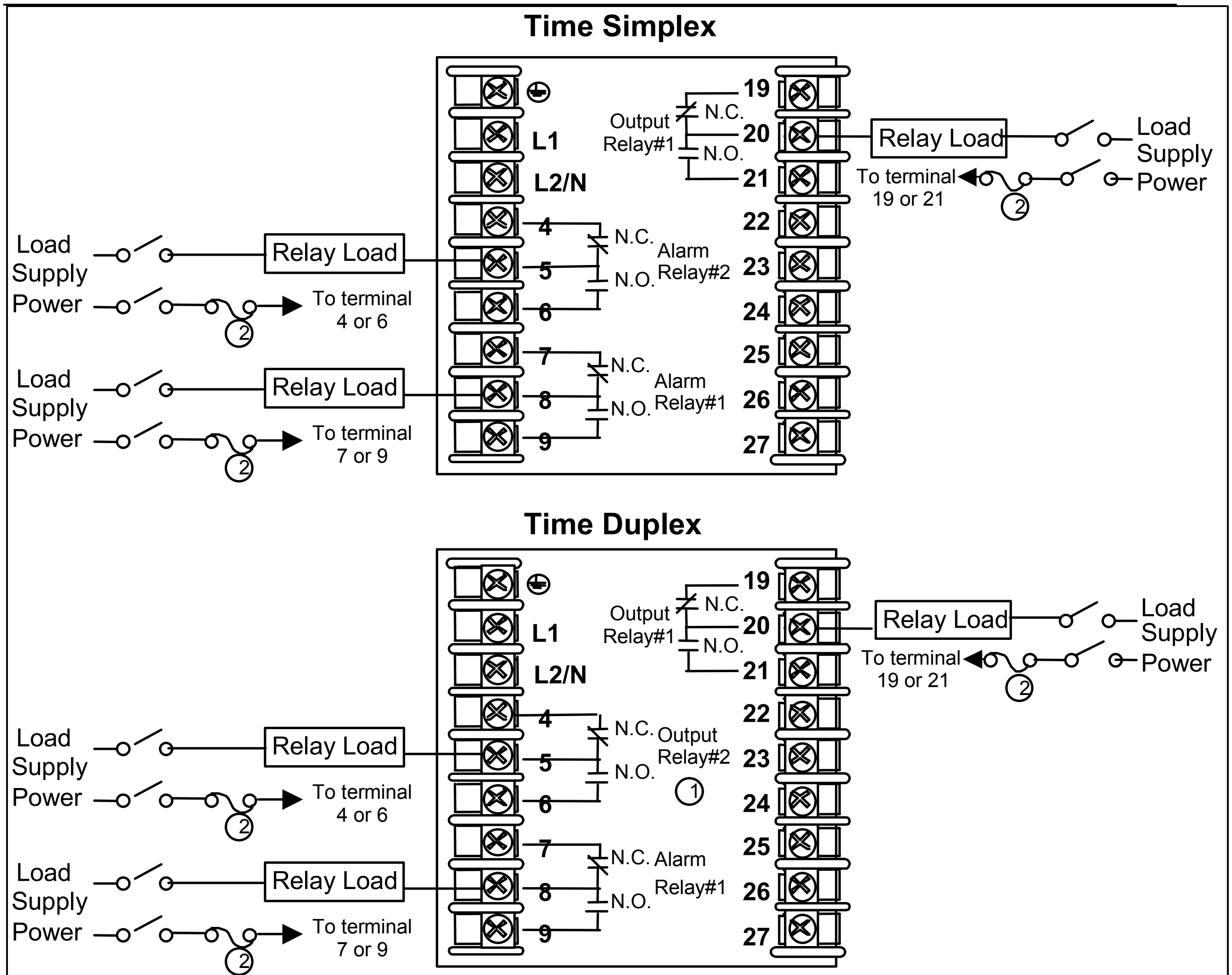
Figure 2-6 Input 1 Connections



1. The 250 ohm resistor for milliamp inputs or the voltage divider for 0-10 Volt inputs are supplied with the controller when those inputs are specified. These items must be installed prior to start up when the controller is wired. For 0-20 mA applications, the resistor should be located at the transmitter terminals if Burnout detection is desired.
2. Splice and tape this junction between the two thermocouples. This junction may be located anywhere between the thermocouples and the instrument terminals, it does not need to be close to the other thermocouple junctions. Both thermocouples must be of the same type. For best accuracy, the two thermocouples should be matched or, preferably, made from the same batch of wire.
3. This controller does not produce a steady current for burnout detection. For that reason, when a thermocouple is used in parallel with another instrument, it may be desirable to configure the burnout selection for this controller to "NOFS" and use the burnout current from the other instrument to also drive this controller.
4. Input 2 is used to measure the Slidewire Input for Position Proportional Control.

xxxx

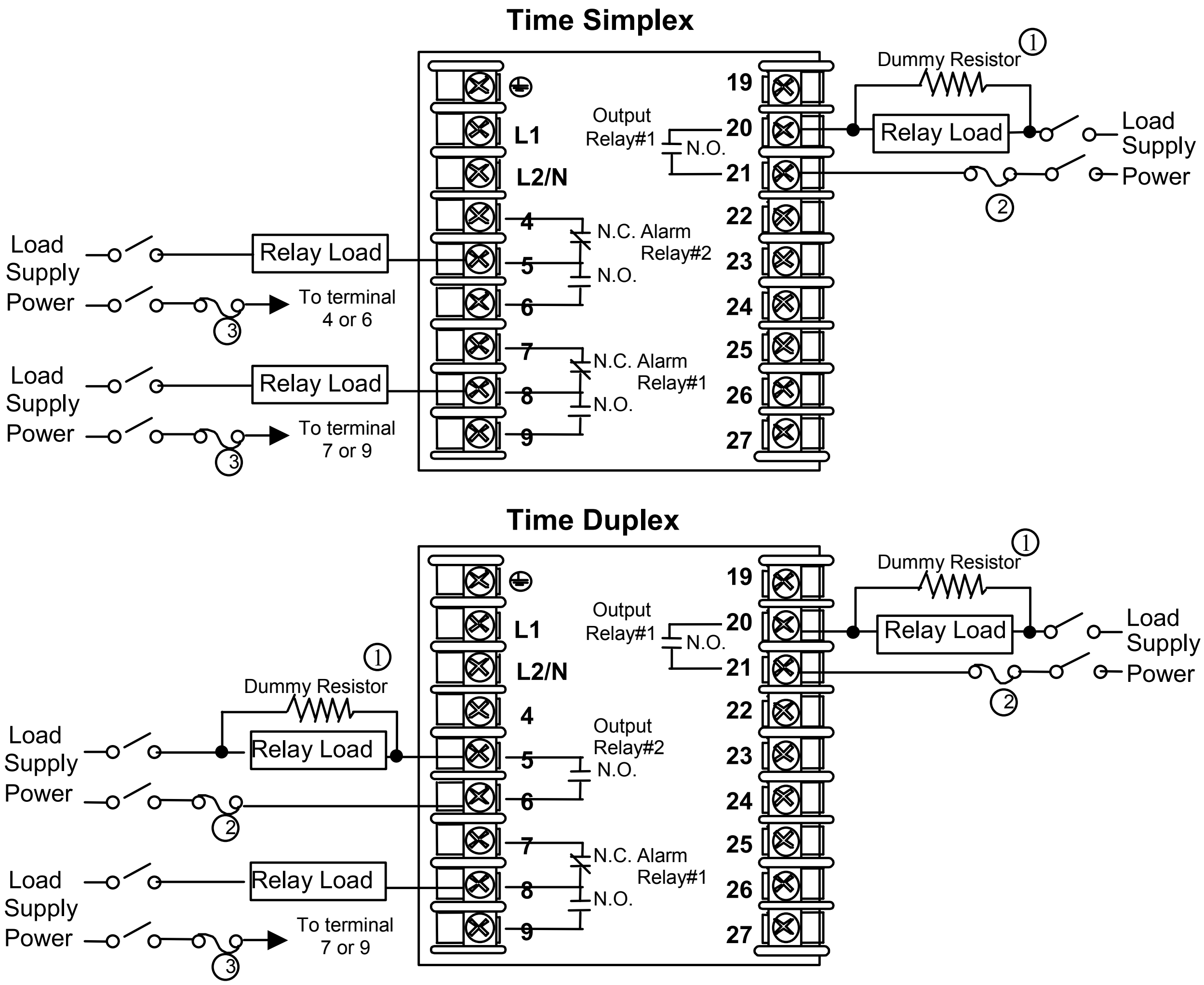
Figure 2-7 Input 2 Connections



1 Alarm #2 is not available with Time Proportional Duplex or Three Position Step Control unless the Dual Relay Option is used.
2 Electromechanical relays are rated at 5 Amps @ 120 Vac or 240 Vac or 30 Vdc.
Customer should size fuses accordingly. Use Fast Blo fuses only.

Figure 2-8 Electromechanical Relay Output

See Table 2-6 for relay terminal connections for other Output Algorithm Types.



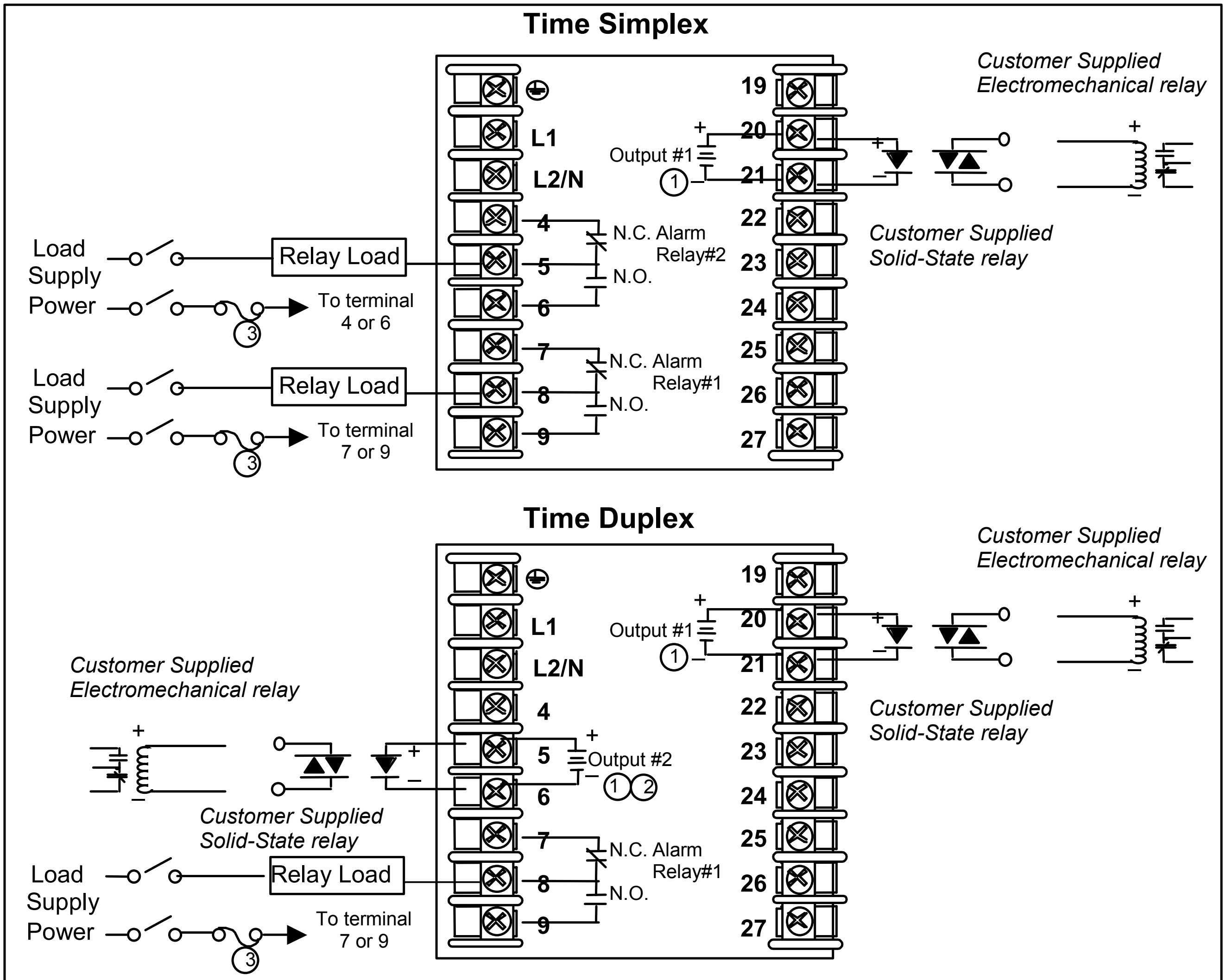
1. If the load current is less than the minimum rated value of 20 mA, then there may be residual voltage across both ends of the load even if the relay is turned off. Use a dummy resistor as shown to counteract this. The total current through the resistor and the the load must exceed 20 mA. Solid State Relays are zero-crossing type.

2. Solid State relays are rated at 1 Amp at 25°C and derated linearly to 0.5 Amp at 55°C. Customer should size fuse accordingly. Use Fast Blo fuses only.

3. Electromechanical relays are rated at 5 Amps @ 120 Vac or 240 Vac or 30 Vdc. Customer should size fuses accordingly. Use Fast Blo fuses only.

Figure 2-9 Solid State Relay Output

See Table 2-6 for relay terminal connections for other Output Algorithm Types.



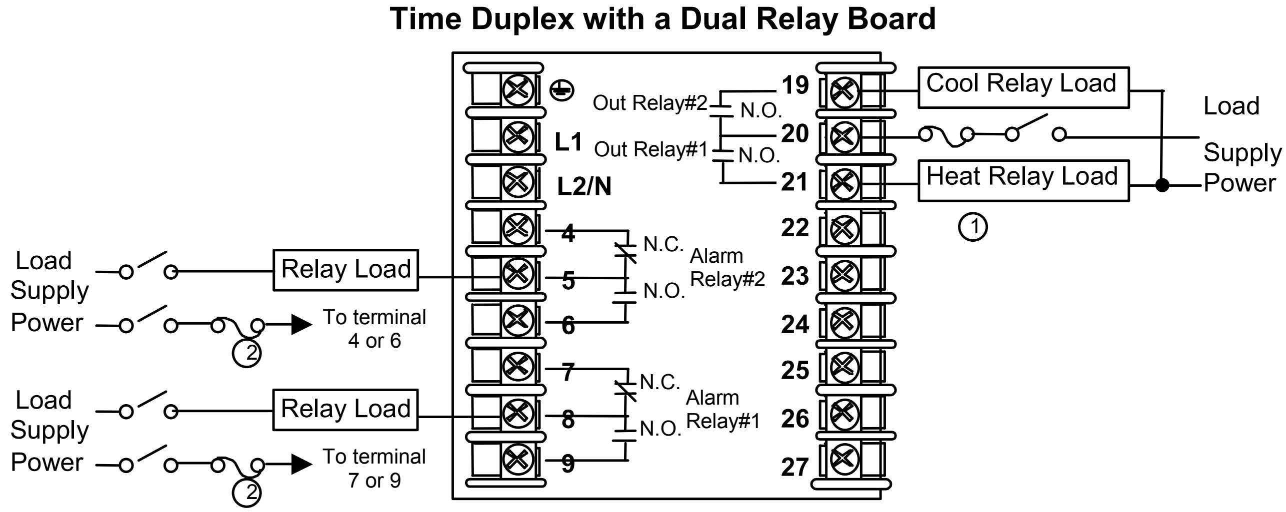
1 **CAUTION** Open collector outputs are internally powered at +30 Vdc. Connecting an external power supply will damage the controller.

2 Alarm #2 is not available with Time Proportional Duplex or Three Position Step Control unless the Dual Relay option is used.

3 Electromechanical relays are rated at 5 Amps @ 120 Vac or 240 Vac or 30 Vdc. Customer should size fuses accordingly. Use Fast Blo fuses only.

Figure 2-10 Open Collector Output

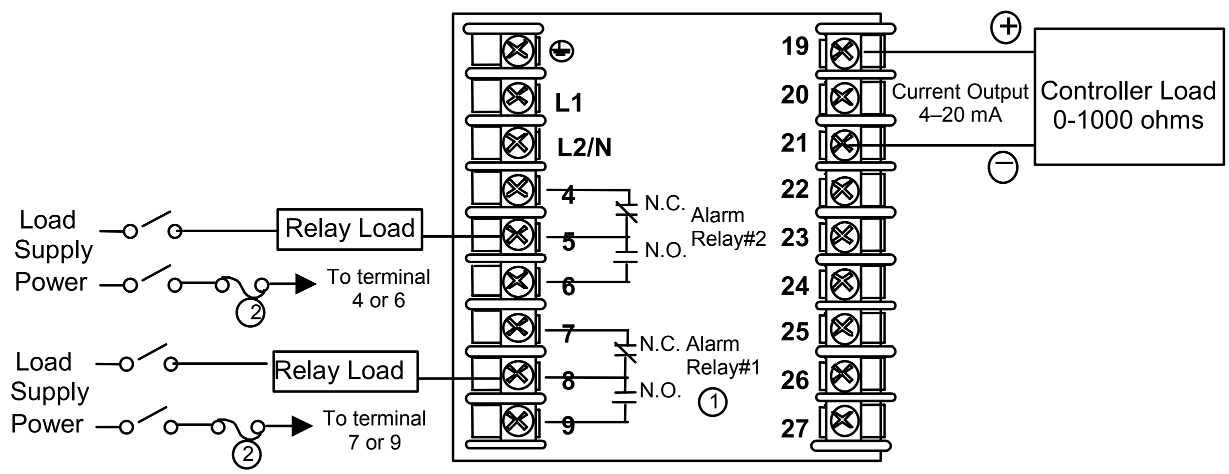
See Table 2-6 for relay terminal connections for other Output Algorithm Types.



1. Dual Electromechanical relays are rated at 2 Amps @120 Vac or 240 Vac or 30 Vdc. Customer should size fuses accordingly. Use Fast Blo fuses only.
2. Electromechanical relays are rated at 5 Amps @120 Vac or 240 Vac or 30 Vdc. Customer should size fuses accordingly. Use Fast Blo fuses only.

Figure 2-11 Dual Electromechanical Relay Option Output

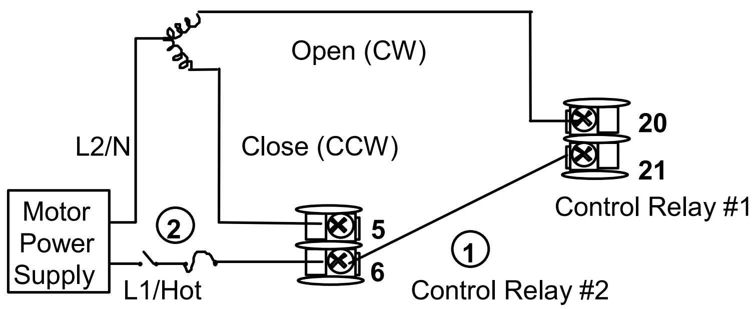
See Table 2-6 for relay terminal connections for other Output Algorithm Types.



1. When the instrument has the Current Output as shown, no Alarms are available when using the Time Proportional Duplex or Three Position Step Control Output Algorithms, as these outputs require both available relays.
2. Electromechanical relays are rated at 5 Amps @120 Vac or 240 Vac or 30 Vdc Customer should size fuses accordingly. Use Fast Blo fuses only.

Figure 2-12 Current Output

See Table 2-6 for relay terminal connections for other Output Algorithm Types.



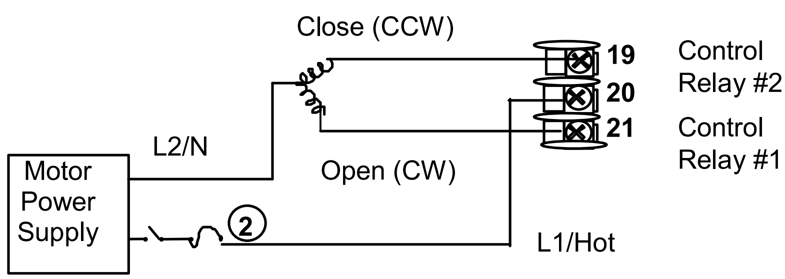
1 Alarm #2 is not available with this configuration.

2 Electromechanical Relays are rated at 5 amps at 120 Vac or 240 Vac or 24 Vdc. Solid State Relays are rated at 1 Amp at 25°C and derated linearly to 0.5 Amps at 55°C. Customer should size fuses accordingly. Use Fast Blo fuses only.

3 See Input 2 Wiring Diagram for Slidewire Connections.

xxxx

Figure 2-13 Position Proportional or Three Position Step Control Connections w/o Dual Relay Option



1 Alarm #2 is available with with this configuration.

2 Dual Electromechanical relays are rated at 2 Amps @120 Vac or 240 Vac or 30 Vdc. Customer should size fuses accordingly. Use Fast Blo fuses only.

3 See Input 2 Wiring Diagram for Slidewire Connections.

xxxx

Figure 2-14 Position Proportional or Three Position Step Control Connections with Dual Relay Option

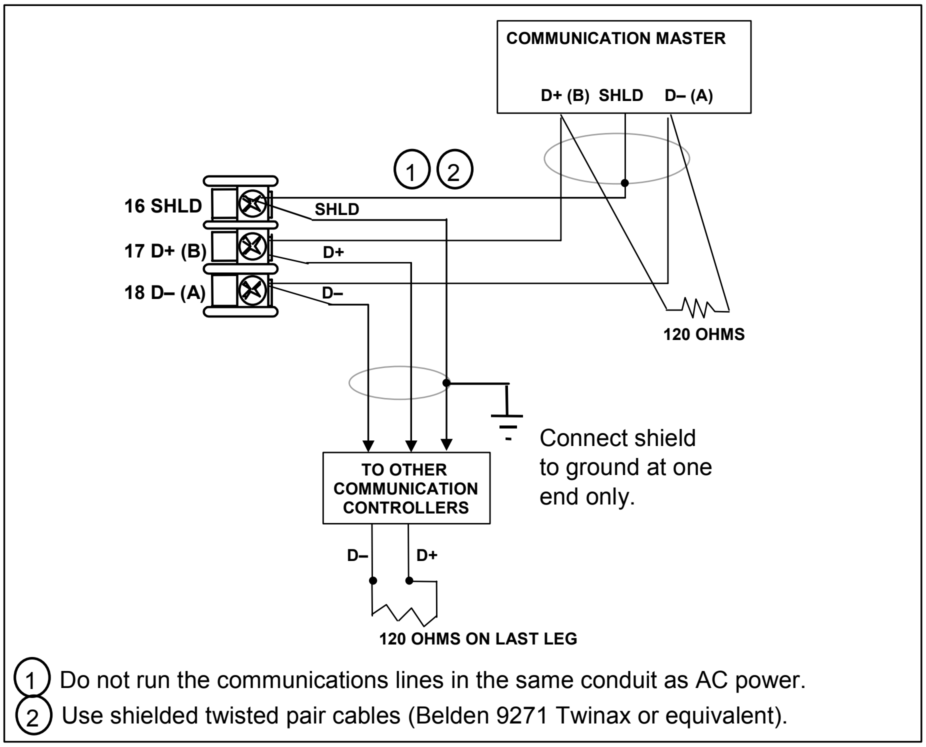


Figure 2-15 RS-422/485 Communications Option Connections

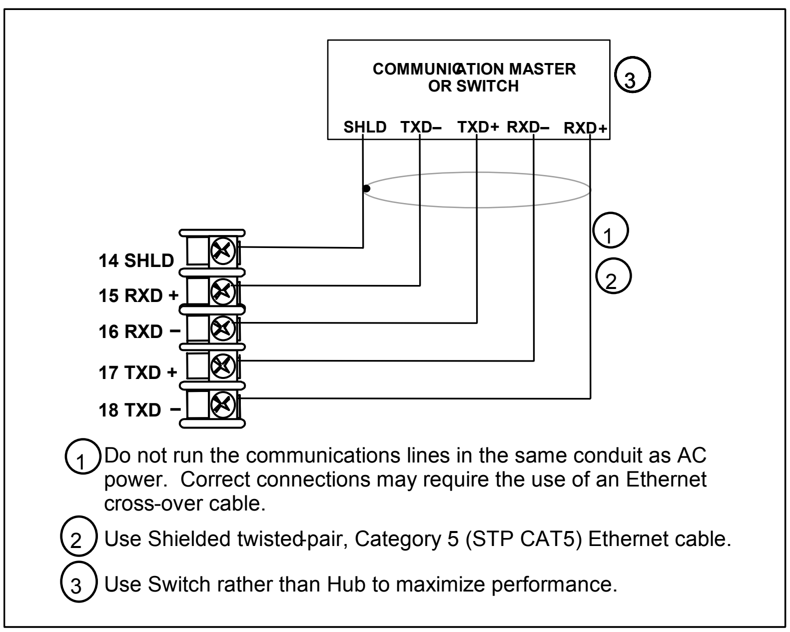


Figure 2-16 Ethernet Communications Option Connections

Figure 2-16 and Table 2-7 shows how to connect a UDC to a MDI Compliant Hub or Switch utilizing a **straight-through cable** or for connecting a UDC to a PC utilizing a **crossover cable**.

Table 2-7 Terminals for connecting a UDC to a MDI Compliant Hub or Switch

UDC Terminal	UDC Signal Name	RJ45 Socket Pin #	Switch Signal Name
Position 14	Shield	Shield	Shield
Position 15	RXD-	6	TXD-
Position 16	RXD+	3	TXD+
Position 17	TXD-	2	RXD-
Position 18	TXD+	1	RXD+

Table 2-8 shows how to connect a UDC directly to a PC utilizing a straight-through cable (wiring the UDC cable this way makes the necessary cross-over connections)

Table 2-8 Terminals for connecting a UDC directly to a PC utilizing a straight-through cable

UDC Terminal	UDC Signal Name	RJ45 Socket Pin #	PC Signal Name
Position 14	Shield	Shield	Shield
Position 15	RXD-	2	TXD-
Position 16	RXD+	1	TXD+
Position 17	TXD-	6	RXD-
Position 18	TXD+	3	RXD+

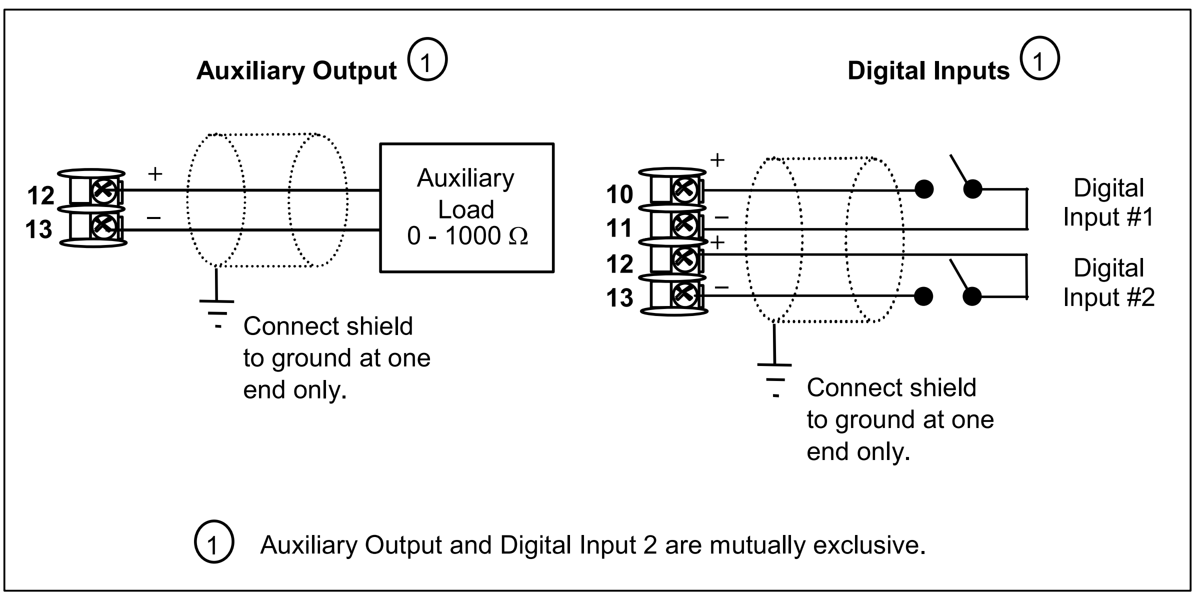


Figure 2-17 Auxiliary Output and Digital Inputs Option Connections

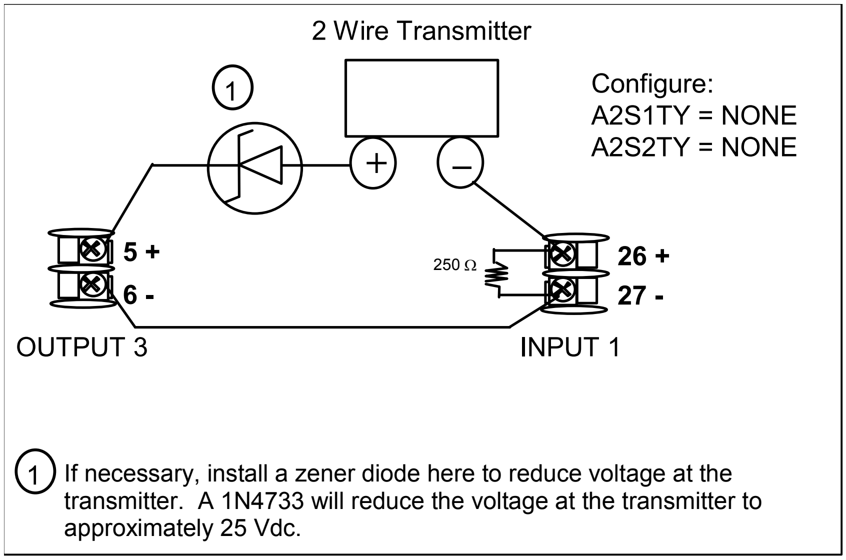


Figure 2-18 Transmitter Power for 4-20 mA — 2 wire Transmitter Using Open Collector Alarm 2 Output

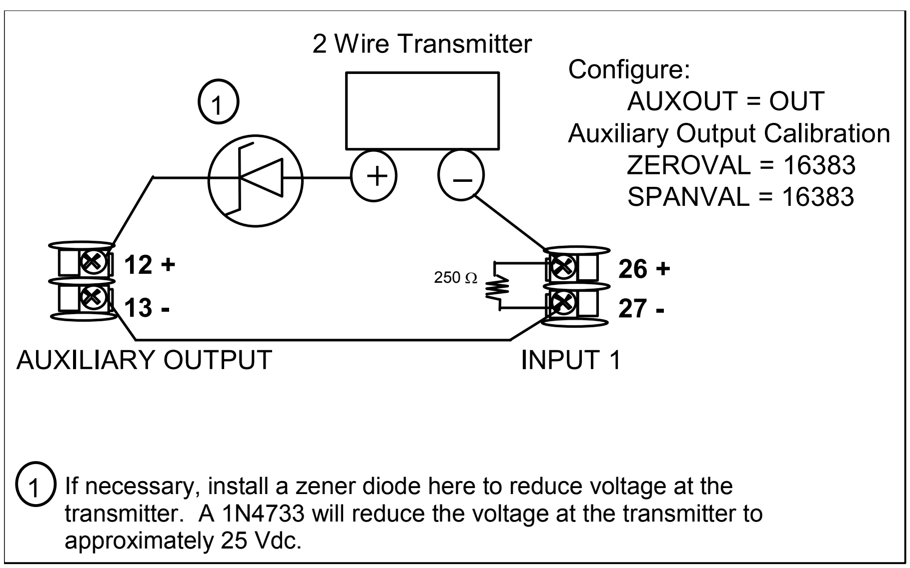


Figure 2-19 Transmitter Power for 4-20 mA — 2 Wire Transmitter Using Auxiliary Output